Xinyi Sun

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EDUCATION

The Hong Kong University of Science and Technology **Hong Kong**
PhD student *Sep 2022 – Present*
Marketing (Advisor: Professor Jia Liu)

National University of Singapore **Singapore**
Msc. Science *Sep 2020 – Dec 2021*
Marketing Analytics and Insights

Renmin University of China **Beijing, China**
BSc. Business *Sep 2016 – July 2020*
Management Science and Engineering

RESEARCH INTERESTS

Topics: Branding, Visual Marketing, AIGC, User Generated Content, Social Media, Advertising
Methodological: Causal Inference, Agentic-LLM, Stable Diffusion, Machine Learning, Field Experiment, Lab Experiment

WORKING PAPERS

* indicates corresponding author

- Xinyi Sun, Jia Liu and Jincheng Fang, ”From Engagement to Equity: Brand Building in Content-Driven Digital Markets”
  ∗ ***Job Market Paper & Dissertation***
- Xinyi Sun, Jia Liu* and Jeehye Christine Kim, ”Curbing Toxicity Without Compromising Privacy? The Effects of Limiting Anonymity on User Engagement” ***Revise and Resubmit (2nd round), Journal of Marketing Research***
- Jia Liu, Qiaoyang Liu and Xinyi Sun*,”Generating Controllable Visual Stimuli: A Guide for Marketing and Social Science Researchers” ***Revise and Resubmit (Major Revision), International Journal of Research in Marketing***
- Song Lin, Zijun (June) Shi and Xinyi Sun, ”How LLM Chatbots Reshape Consumer Search: Evidence on Attention Allocation” [Link] *Manuscript under preparation*
- Wenbo Wang, Jia Liu and Xinyi Sun*, ”Integrating General AI into Vertical Domains: A Theory- and Expertise-Driven Framework from Narrative Content Marketing” *Manuscript under preparation*

SELECTED WORK IN PROGRESS

- Xinyi Sun and Jia Liu, “Generative Ad Design and Brand Personality.” *Method development in progress*
- Xinyi Sun and Jia Liu, “Branding Through Visual Content: Cross-Cultural Evidence from Digital Content Markets” *Method development in progress*

CONFERENCE PRESENTATION

Asia-Pacific Marketing Academy Annual Conference *Guangzhou, Sep 2023*

"User ID Format on User Generated Toxic Content on Social Media Platforms." *Doctoral Forum*

4th Artificial Intelligence in Management Conference *Los Angeles, Mar 2024*
"Towards Intelligent Shopping Assistant: Can LLM Chatbot Empower Consumer Decision Making?"

2025 Conference on AI, ML, and Business Analytics *New York, Dec 2025*
"Generating Controllable Visual Stimuli: A Guide for Marketing and Social Science Researchers"

2026 ISMS Marketing Science Conference *Lisbon, June 2026*
"Curbing Toxicity Without Compromising Privacy? The Effects of Limiting Anonymity on User Engagement"

COURSES

Seminar in Quantitative Modelling, I & II
Experimental Design for Behavioral Research
Current Topics in Consumer Research
Behavioral Decision Theory
Mathematics for Business and Economics
Micro-economic Theory
Econometrics
Empirical Industrial Organization
Topics in the Economics of IT
Human-Computer Interaction
Machine Learning

TEACHING & SERVICE

Teaching Assistant
Navigating Business Decisions with Data and Causality (Master) *Fall 2023,2024*

Guest Lecturer
Marketing Management (Undergraduate) *Spring, Fall 2024*

Guest Lecturer
Brand Management (Undergraduate) *Spring 2025*

INTERNSHIP

ByteDance, Management Analyst Intern *2020*

Amazon Web Service, Field Marketing Intern *2019, 2020*

AWARDS AND COMPETITIONS

- **Fellow, ISMS Marketing Science Doctoral Consortium**, 2026
- **Young Scientist Scholarship** HKUST, 2024 – 2025
- **UGC Research Travel Award**×4, HKUST, 2023, 2024, 2025, 2026
- **Postgraduate Studentship** HKUST, 2022 – Present
- **University-level scholarship (top 10%)** Ministry of Education, China, 2019

REFERENCES

Jia Liu (Advisor)
Associate Professor of Marketing
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J. Christine Kim
Assistant Professor of Commerce
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RESEARCH ABSTRACTS

"**From Engagement to Equity: Brand Building in Content-Driven Digital Markets.**" With Jia Liu and Jincheng Fang. *Job Market Paper & Dissertation*

Abstract: Digital platforms make brand engagement highly visible, but they also make it harder to know whether visible audience response reflects durable brand building or transient attention. We study this question in TikTok Shop, an integrated content-commerce platform where brands continuously produce video content and consumers can engage and purchase within the same ecosystem. Using 357,418 brand-month observations from 21,727 shops across Vietnam, Thailand, Indonesia and the United States, we recover sales-based brand equity (SBBE) from a structural demand model and develop two measures needed to study digital brand building at scale. First, we introduce Consensus-based Rubric Agentic Framework for Theory-grounded measurement (CRAFT), a multi-agentic LLM measurement architecture that converts unstructured videos into validated measures of content-embedded branding strategy, i.e., differentiation, relevance, esteem, and knowledge. Second, we construct engagement momentum, the change in accumulated engagement capability, to distinguish durable consumer responsiveness from short-term engagement flow. The evidence shows that SBBE remains a persistent marketplace asset in content-marketing contexts. Engagement momentum, rather than contemporaneous flow, is the digital interaction signal most closely associated with SBBE accumulation. Content-embedded branding strategy operates configurationally: relevance-oriented content is penalized unless paired with sufficient differentiation. The momentum-equity link is strongest in markets where brand positions are still forming. The findings show that digital brand building depends on strategic meaning and sustained audience responsiveness, not content activity or engagement volume alone.

"**Curbing Toxicity Without Compromising Privacy? The Effects of Limiting Anonymity on User Engagement**" With Jia Liu and Christine Kim. *Revise and Resubmit (2nd round), Journal of Marketing Research*

Abstract: As toxic behavior escalates online, platforms face a persistent dilemma: how to curb undesirable content without relying on intrusive identity disclosure or heavy-handed moderation. We study whether and how subtle reductions in anonymity can improve discourse quality by exploiting a sharp policy change on a large university forum where users could post under no ID or under persistent, non-personal IDs. The intervention restricted fully anonymous posting from unlimited to three posts per day. Using a Regression Discontinuity in Time (RDiT) design, we estimate the causal impact of this change on a multidimensional set of content outcomes—including negative and positive emotion, toxicity, mental harm, and informativeness—measured with LIWC dictionaries and human-validated large-language-model labels. The intervention produced sizable and immediate improvements in discourse quality. Toxic, mentally harmful, and negative posts fell by 20-50%, while informative and positive posts increased. Importantly, these quality gains occurred with only modest reductions in posting volume and no statistically significant decline in overall commenting activity. Heterogeneous treatment effects and a preregistered randomized experiment converge on a common mechanism—perceived traceability. Users who felt more recognizable (e.g., through distinctive IDs) exhibited clearer behavioral recalibration. Our findings demonstrate that persistent, non-personal identifiers can meaningfully improve online discourse, offering a light-touch alternative to real-name policies or heavy-handed moderation.

"**Generating Controllable Visual Stimuli: A Guide for Marketing and Social Science Researchers**" With Jia Liu and Qiaoyang Liu. *Revise and Resubmit (Major Revision), International Journal of Research in Marketing*

Abstract: Visual stimuli, such as human faces and product presentation, are pivotal in studying human perception, judgment, and decision-making. Traditional methods for creating experimental stimuli face significant challenges, such as difficulties in isolating specific visual features and limited scalability/generalizability. While recent advancements in computer vision have produced cutting-edge AI image generation technologies with great potential to address these challenges, there is a lack of understanding about what these tools are and how they can be utilized for marketing and social science research (due to high requirements for domain knowledge and usage experience). This paper bridges the gap by providing research-focused, practical guidance on Stable Diffusion, which can generate high-fidelity, controllable visual stimuli in a scalable and cost-effective manner. This paper first explains the key concepts and techniques of Stable Diffusion, and then showcases two sets of applications, with interactive and code-free demos available on the accompanying website. These applications involve systematic manipulations of facial features and product-related images, some of which replicate the stimuli used in existing literature. Through these applications, this paper demonstrates the potential of Stable Diffusion to unlock significant research opportunities in visual perception and decision-making.

"**How LLM Chatbots Reshape Consumer Search: Evidence on Attention Allocation**" With Song Lin and Zijun (June) Shi. [Link]

Abstract: Generative AI chatbots are widely expected to accelerate consumer search by lowering the cost of accessing information. This paper challenges that intuition. Using an incentive-aligned experiment on a purpose-built e-commerce platform, we find that LLM-powered chatbot assistance increases, rather than reduces, total search time. This effect is driven by deeper inspection of individual products, not broader exploration. We argue and show that this pattern reflects a fundamental shift in the search mechanism. While classical search theory has emphasized the cost of *accessing* information, the cognitive cost of *allocating* attention across attributes has received less scrutiny. The chatbot reduces the cost of dynamically updating attribute importance, shifting consumers from batch to adaptive attention. A follow-up experiment confirms that chatbot users significantly reweight their preferences, and an attention-priming task partially substitutes for the chatbot's effect, providing causal evidence for the mechanism. These findings suggest that as AI drives information-access costs toward zero, the binding constraint on consumer search is shifting from access to attention allocation. We discuss implications for search theory and the design of AI-assisted choice environments.

"**Integrating General AI into Vertical Domains: A Theory- and Expertise-Driven Framework from Narrative Content Marketing**" With Wenbo Wang and Jia Liu.

Abstract: Large language models (LLMs) offer strong potential for specialized domains, yet their effectiveness is limited by the difficulty of incorporating tacit expert knowledge. This paper addresses this "vertical integration gap" by developing and validating a structured, domain-informed prompting framework that embeds expert knowledge into LLM workflows. Using interviews with professional content creators, we codify fifteen evaluation dimensions and organize them into a four-stage prompting framework that mirrors expert cognition. Empirical results with experts and consumers on REDnote show that our approach replicates expert–AI collaboration performance, achieving comparable quality while reducing production time and improving consumer engagement relative to top human-generated content. Field deployment further indicates substantial business impact, including growth in output, engagement, and revenue with stable labor costs. Beyond content marketing, we derive generalizable principles for integrating domain expertise into generative AI, demonstrating how structured prompting can transform general-purpose models into scalable, domain-specific systems.